

**Zeyu Wang**

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**EDUCATION**

University of California, Los Angeles  
 Master of Computer Engineering  
 (Master Degree Conferred Date: Dec 18, 2020)

Sept 2019 ~ Dec 2020

University of California, Los Angeles  
 PhD program of Computer Engineering  
 (Conducted Research, not obtained PhD degree)

Dec 2020 ~ June 2022

**RESEARCH AREA**

Machine Learning, Artificial Intelligence

Large Language Model

**PROFESSIONAL EXPERIENCE**

Machine Learning Engineer, **Amazon, Inc**  
 Researcher of Language Understanding Synthesis Lab of **UCLA**

2022 ~ 2024

2024 ~ Present

**NOTABLE PROJECTS****Language Understanding & Synthesis Lab of UCLA**

**LLM Causal Reasoning: Developed fundamental causal reasoning evaluation benchmark to assess LLMs causal reasoning capability. (published paper for ACL 2024 )**

- Causal Reasoning Benchmark Development: Designed and curated a comprehensive dataset aimed at evaluating the causal reasoning ability of large language models (LLMs)
- Causal Inference Engine: Developed a Causal Inference Engine to automatically generate causal hypotheses and label pairs based on predefined causal relationships , ensuring high- quality, diverse causal scenarios for testing
- Experimental Evaluation Framework: Conducted extensive evaluations across multiple types of causal queries to measure LLM performance, identifying key limitations in current models and setting a new standard for causal reasoning assessment in NLP research

**AI Agent Development and Fine-Tuning for Intelligent Automation**

- Developed and fine-tuned **AI agents** for complex decision-making tasks, leveraging **large language models (LLMs)** and adaptive reasoning techniques to enhance automation capabilities.
- Implemented **Function Calling** mechanisms, enabling AI agents to dynamically interact with external tools, APIs, and structured data sources, improving task execution efficiency.
- Designed a **scalable fine-tuning pipeline**, optimizing model performance across diverse domains using **supervised fine-tuning, reinforcement learning, and prompt engineering**.
- Integrated **multi-modal AI capabilities**, allowing agents to process and generate **text, speech, image, and video outputs**, ensuring seamless interoperability across different applications.
- Built AI-driven **evaluation and execution platforms**, providing **real-time monitoring, benchmarking, and optimization** for agent performance in production environments.
- Deployed AI agents using frameworks such as **LangChain and XAgent**, implementing modular architectures that allow adaptive behavior, self-learning, and efficient task orchestration.
- Optimized **low-latency inference pipelines**, reducing computational overhead by **25%** while maintaining high accuracy, ensuring real-time responsiveness in mission-critical applications.

### **Multi-Stage Ensemble Architecture for Enhanced Logical Reasoning in LLMs (Accepted)**

- Developed a novel cascade network architecture combining dynamic candidate filtering, adaptive self-attention, and ensemble decision voting to improve LLMs' logical reasoning capabilities
- Implemented confidence-based candidate selection and multi-head adaptive attention mechanisms to dynamically adjust attention weights, enabling more precise logical inference and connection identification
- Enhanced model performance through an intermediate logical representation layer and ensemble decision strategy, improving accuracy from 81.48% (base model) to 88.86% (full ensemble) while maintaining efficient inference time
- Leveraged parallel processing via multiple specialized components: candidate filtering for relevance scoring, adaptive attention for dynamic weighting, and voting mechanisms for robust decision making, resulting in superior performance on complex multi-step reasoning tasks

### **Amazon, Inc**

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#### **AI Agent for Intelligent Risk Detection and Prevention**

- Developed and deployed an AI-powered agent for real-time risk detection and decision automation, integrating LLMs and anomaly detection for enhanced accuracy.
  - Designed and optimized data preprocessing pipelines, feature engineering techniques, and model training workflows, leveraging supervised learning and self-adaptive AI agent strategies.
  - Implemented Function Calling to enable AI agents to dynamically interact with external data sources and APIs, improving real-time adaptability in risk assessment.
  - Deployed the AI agent using AWS (S3 for storage, Lambda for compute, SageMaker for model deployment), ensuring scalability, efficiency, and robust model lifecycle management.
  - Automated agent retraining and self-improvement mechanisms, enabling continuous learning from new patterns and reducing false positives in risk assessments.
- Collaborated with cross-functional teams to integrate the AI agent into production environments, achieving a 20% improvement in response time and increasing decision accuracy.

#### **Real-Time AI-Driven Dynamic Pricing and Bidding System**

- Developed a real-time dynamic pricing engine using reinforcement learning and time-series forecasting (LSTM, Transformer) to optimize price adjustments across millions of products.
- Designed a low-latency bid optimization model for Amazon Advertising, leveraging multi-armed bandits and deep Q-learning to dynamically adjust ad bids, increasing ROI by 12%.
- Built a millisecond-level competitive pricing system, integrating real-time competitor price tracking and demand forecasting to maximize revenue efficiency.
- Optimized pricing decision latency using AWS Lambda, Redis caching, and ONNX-accelerated inference, reducing price update delays by 30% while handling millions of requests per second.
- Developed an anomaly detection system using Graph Neural Networks (GNNs) and Isolation Forests, reducing pricing errors and fraudulent activities by 25%.
- Deployed the system using AWS SageMaker, Kinesis, and EC2, supporting billions of daily updates with 99.99% uptime and automated scalability.

### **Reviewer Experience**

NeruIPS 2024

ACM Transactions on Computing for Healthcare

European Journal of Agronomy

Advances in Continuous and Discrete Models

### **AWARDS**

- 2017 China National Robot Competition Second Class Prize
- 2017 China Intelligent Car Competition, Second Class Prize

## NOTABLE PUBLICATIONS

[CausalBench: A Comprehensive Benchmark for Evaluating Causal Reasoning Capabilities of Large Language Models](#)

Zeyu Wang (ACL 2024 Published)

[Optimizing StockMarket Return Forecasts with Uncertainty Sentiment: Leveraging LLM-based Insights](#)

Qian Zhang&Zeyu Wang, Xuerong Shang, Jiarui Rao

(2024 6th International Conference on Machine Learning, Big Data and Business Intelligence Accepted)

[A Dynamic Adaptive Data Filtering and Sampling Framework for Optimized Fine-Tuning of Large Language Models](#)

Zeyu Wang et, al (2025 5<sup>th</sup> International Conference on Consumer Electronics and Computer Engineering Accepted)

[Enhancing Logical Reasoning in Large Language Models via Multi-Stage Ensemble Architecture with Adaptive Attention and Decision Voting](#)

Zeyu Wang (ICML 2025 under review)

[A Hybrid Framework for Enhanced Prompt Generation: Integrating Mean Templates, Fine-Tuned Models, and Clustering Mechanisms](#)

Zeyu Wang et,al (NNICE 2025 accepted)